

OCTAVIO MOSQUEDA

Full Stack Developer & AI Engineer | Node.js · Next.js · Go · Azure AI · Python

octaviomosq26@gmail.com | [Portfolio](#) | [GitHub](#) | English C1 · Spanish Native

Software Engineer with professional experience building end-to-end web applications and AI-powered pipelines. Proficient in JavaScript/TypeScript (Next.js, Node.js), Go, and Python; applied across real production systems: a HIPAA-aligned medical SaaS with Azure AI Document Intelligence, a concurrent booking backend in Go, and a cloud-native staffing platform.

PROFESSIONAL EXPERIENCE

Truckers Talent INC · Full Stack Developer Intern January 2026 – May 2026

- **Payment & Database Engineering:** Architected a secure Stripe subscription system and optimized PostgreSQL schemas supporting multi-tier subscription logic for recurring billing consistency.
- **Automated Systems & Search Optimization:** Developed a Nodemailer-based email engine and implemented advanced geolocation search filters with Next.js for real-time alerts and user notification modules.
- **Cloud Infrastructure Management:** Orchestrated cloud storage and hosting solutions using AWS (S3) and DigitalOcean, optimizing media asset delivery and maximizing application uptime.

KEY PROJECTS

Thyroidea Full-stack Medical SaaS · Azure AI · Node.js · React · Supabase · PostgreSQL

- **OCR Pipeline:** Built an end-to-end OCR pipeline using Azure Document Intelligence (prebuilt-layout) that extracts 9 thyroid markers (TSH, FT4, FT3, AntiTPO, VitaminD, Ferritin and others) from unstructured lab PDFs. Implemented a domain-specific anchor-value parser with bilingual keyword mapping (ES/EN), automatic pmol/L → ng/dL unit normalization, and per-user daily rate limiting.
- **Security & HIPAA Alignment:** Built custom Express middleware stack — JWT auth, RBAC, resource ownership validation and startup route introspection. Enforced HIPAA-aligned audit logging, pg_cron-based log retention, and raw OCR text gating. File names masked in logs; documents never persisted to disk.
- **Billing & Notifications:** Implemented full subscription lifecycle via Lemon Squeezy (HMAC-SHA256 webhook verification, cancellation/recovery flows) and a server-side push scheduler with deduplication for reminders.

Go Concurrency Booking Backend SaaS · Go · Redis · PostgreSQL · Docker · Kubernetes

- **Concurrency & Integrity:** Eliminated race conditions under concurrent load via Redis distributed locks and PostgreSQL optimistic locking, ensuring zero double-bookings across 100+ simultaneous requests.
- **Infrastructure:** Built JWT auth, atomic Lua rate limiter in Redis, and a cleanup worker with panic recovery; integrated Kubernetes health probes and graceful shutdown.

TECHNICAL SKILLS

Web Ecosystem: JavaScript (ES6+), TypeScript, React, Next.js, Vite, Node.js, Express.js, Tailwind CSS, i18next

Backend & Systems: Python, Go (Golang), Java, PHP, Django, Bash, PowerShell

Databases & Data: PostgreSQL, MySQL, MongoDB, Supabase (RLS + pg_cron)

Cloud & DevOps: AWS (S3/CLI), DigitalOcean, Docker, Kubernetes, Git, Linux, Postman

AI & Specialized: Azure AI (Document Intelligence), NLP, Machine Learning, Stripe / Lemon Squeezy APIs

EDUCATION & CERTIFICATIONS

Universidad Nacional del Sur · Information Systems Engineer 2024 – Present

Certifications: Meta — Advanced React.js · Programming with Python · Full-Stack Developer · Version Control | IBM — Artificial Intelligence Technologies